Analysis output

My data analysis contained the followimng stages

1. Data Pre-processing: Preparing the raw data for analysis by cleaning, transforming, and restructuring it.
2. Outlier Detection: Identifying and addressing data points that deviate significantly from the norm.
3. Missing Value Handling: Implementing strategies to address gaps or incomplete information within the dataset.
4. Hypothesis Testing: Employing statistical methods to test assumptions and draw conclusions from the analyzed data.
5. Data Visualization: Creating visual representations (charts, graphs) to effectively communicate insights derived from the analysis

Data_Manipulation_Process.R mcar_result

Filter

	statistic	df	p.value	missing.patterns
1	1809.34	1102	0	54

Figure 1 (Little MCAR Test)

Analysis of the provided figure indicates a statistically significant result. The calculated test statistic of df (1102) with 1102 degrees of freedom yielded a p-value of 0. This value is significantly less than the conventional threshold of 0.05, so leading to the rejection of the null hypothesis (H_0). so we conclude that the missing values in the dataset are not Missing Completely at Random (MCAR). This implies that the pattern of missingness is likely related to other observed or unobserved variables within the data, suggesting either Missing at Random (MAR) or Missing Not at Random (MNAR) mechanisms. There we can conclude that a more advanced imputation technique, specifically multiple imputation, is appropriate for handling the missing values effectively.

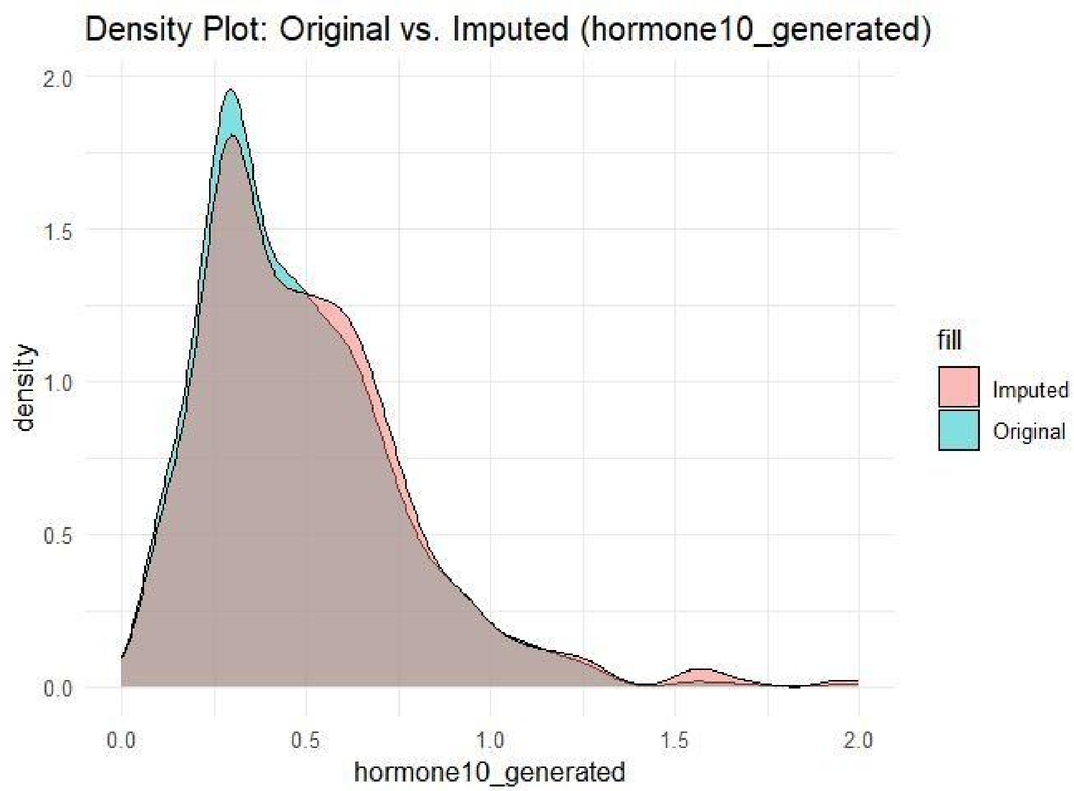

Figure 2 : Hormone Density Plot

We can see that the Density Plot illustrates the distribution of the variable "hormone10_generated" before and after addressing missing values. sThe blue region represents the initial data distribution, while the brown region depicts the distribution following imputation (filling in missing values). The near-identical shapes of these distributions indicate that the imputation method effectively preserved the original data pattern and spread. This suggests a successful imputation process, as the artificial values introduced did not unduly distort the underlying data structure

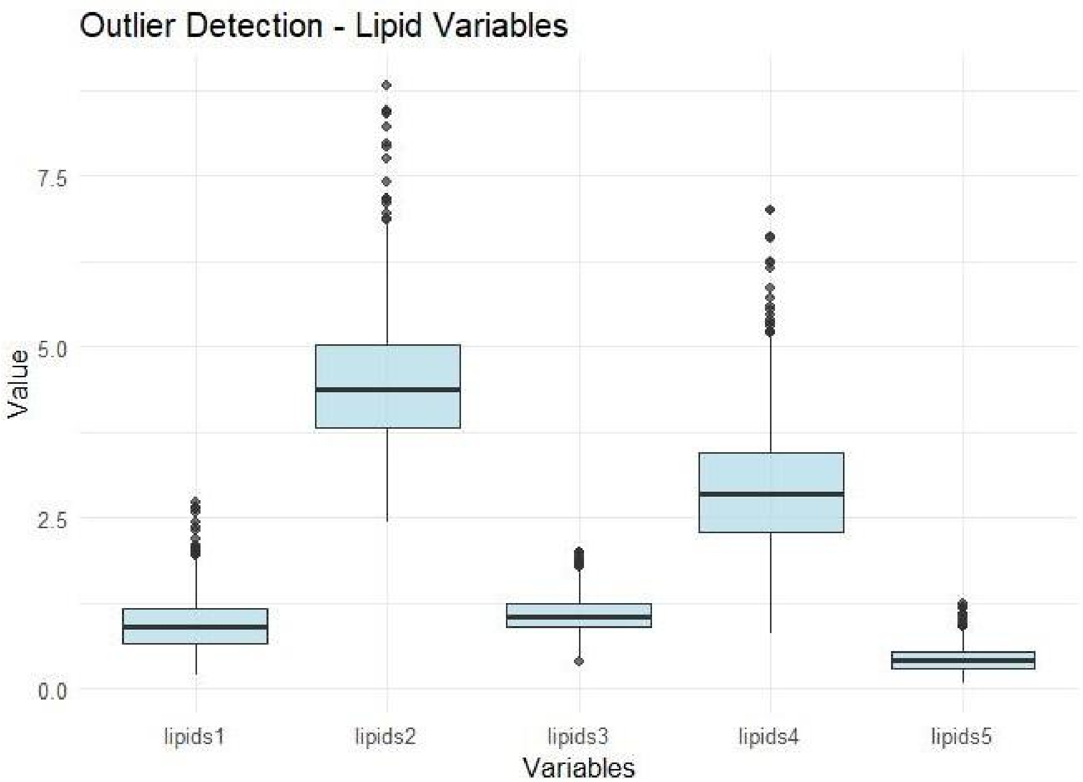

Figure 3 : Outlier Detection boxplot

The figure above presents a comparative analysis of five lipid measurements (lipids1 through lipids5). Outlier values, represented by small circles positioned above and below the boxes, indicate data points that deviate significantly from the general distribution. Notably, lipids2 and lipids4 exhibit the greatest number and magnitude of outliers, suggesting a higher prevalence of extreme values within these specific lipid measurements.

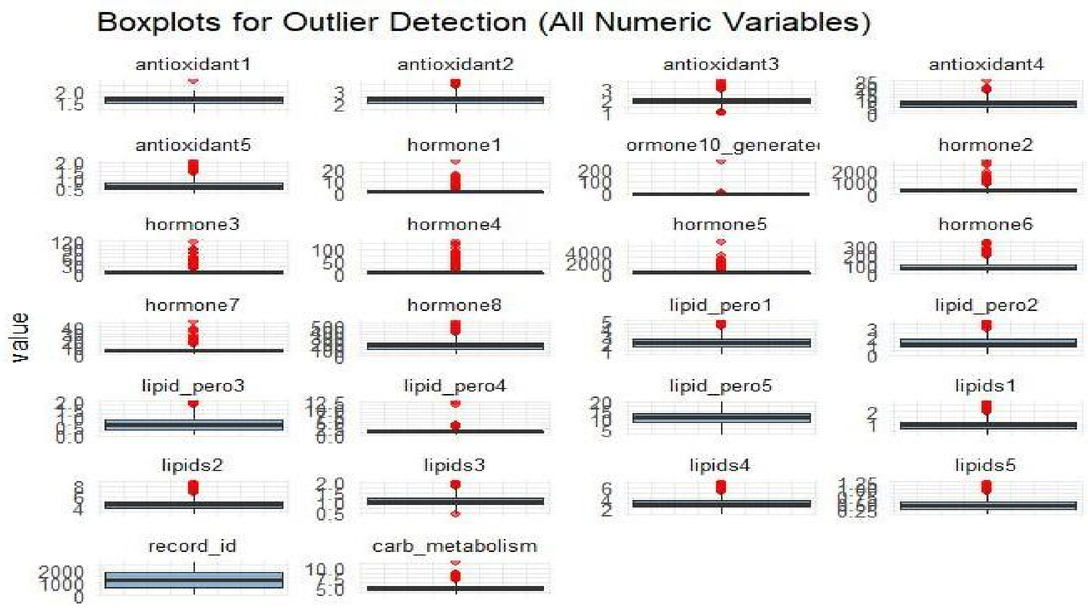

Figure 4 : T h e s Boxplots for Outlier Detection

We can see from the figure above reveals potential outlier values within each numerical column of the dataset. The figure illustrates the distributions of various variables, including antioxidant levels (antioxidant1-5), hormone concentrations (hormone1-8), lipid peroxidation markers (lipid_pero1-5), and lipid profiles (lipids1-5). The outliers, represented by red dots, deviate significantly from the typical range of values observed for each variable. Notably, a substantial number of outliers are present in the hormone and lipid measurements, suggesting the presence of numerous extreme values across the dataset.

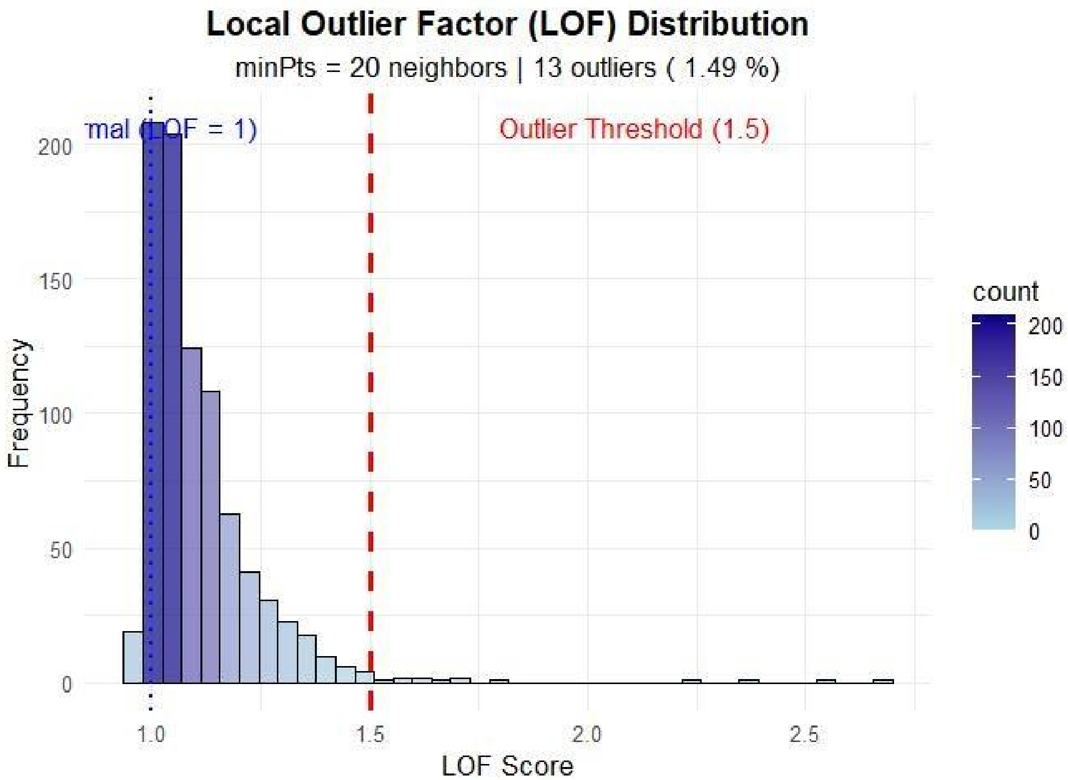

Figure 5 : The Histogram for Outlier Detection

The histogram depicting the Local Outlier Factor (LOF) scores reveals a predominantly normal data distribution. Scores clustering around 1 indicate typical data points. The red dashed line, signifying the outlier threshold of 1.5, identifies points exceeding this value as outliers. Employing 20 neighbors in the LOF calculation resulted in the identification of 13 outliers, representing a mere 1.49% of the total dataset. This suggests that the majority of the data exhibits consistency and normalcy, with only a negligible proportion comprising extreme or anomalous values.

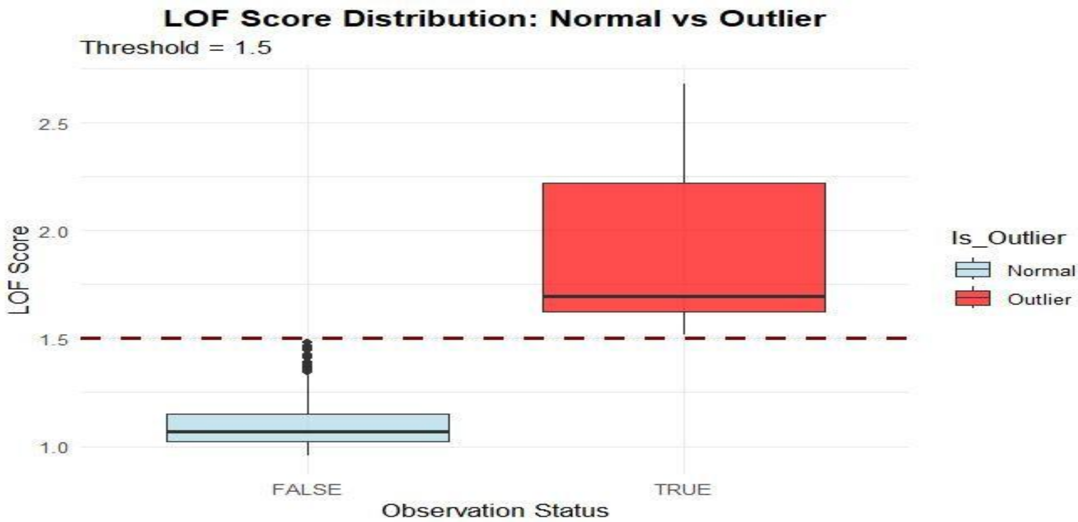

Figure 6 : Boxplots comparing LOF scores

The boxplot illustrates a comparison of Local Outlier Factor (LOF) scores for normal data points versus outlier data points. Normal data points, represented on the left side (FALSE), exhibit LOF scores consistently below the predefined threshold of 1.5 (denoted by the red dashed line). This indicates their conformity with the general data pattern and lack of unusual characteristics. Conversely, outlier data points, depicted on the right side (TRUE), display LOF scores significantly exceeding the threshold, confirming their deviation from the norm. The distinct separation between the two groups strongly suggests the effectiveness of the employed method in accurately identifying outliers within the dataset.

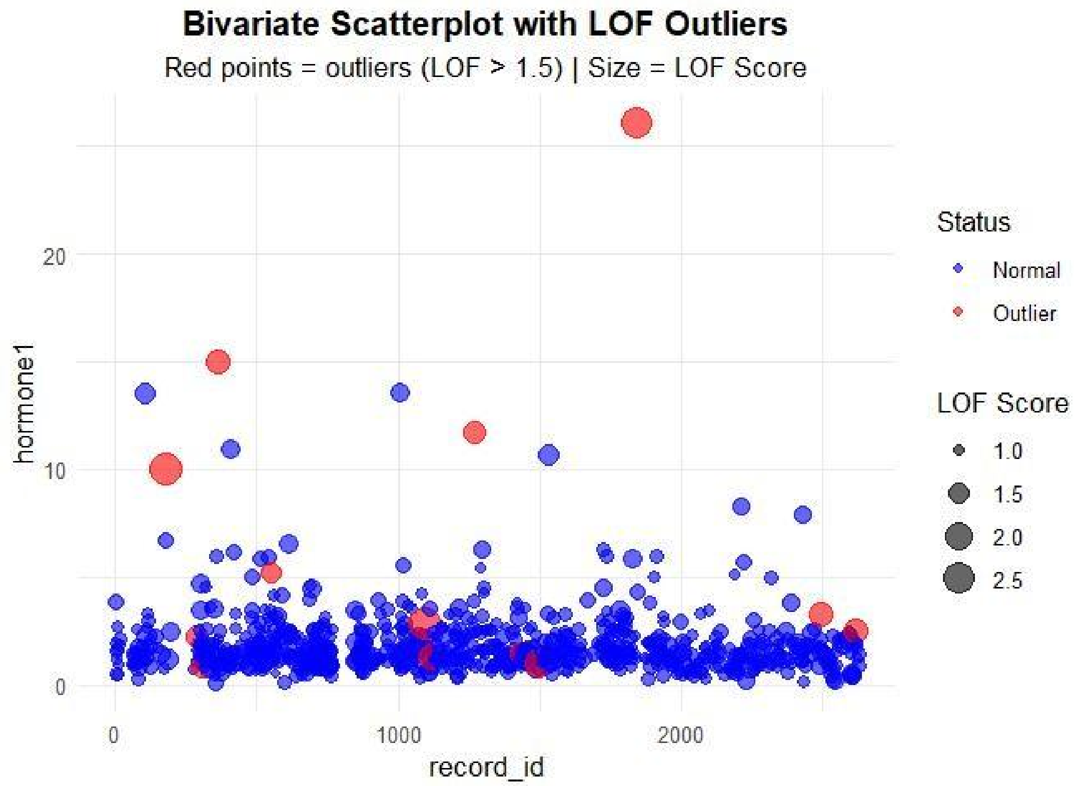

Figure 7: Bivariate Scatterolot

The scatterplot illustration reveals hormone1 levels distributed across various record identifiers. Normal observations are represented by blue points, while red points signify outliers detected using the Local Outlier Factor (LOF) algorithm, exhibiting scores exceeding 1.5. A discernible pattern emerges: the majority of data points congregate in a dense cluster near the lower end of the hormone1 scale (values ranging from approximately 0 to 5). Notably, several red points deviate significantly upwards, indicating anomalous spikes in hormone levels. The intensity of the red color corresponds to the magnitude of outlier status, with larger dots representing stronger deviations.

Conclusion

It can be concluded that the missing data mechanism is classified as either Missing at Random (MAR) or Missing Not at Random (MNAR), rather than Missing Completely at Random (MCAR). Furthermore, Predictive Mean Matching (PMM) imputation was successfully implemented with high accuracy. Multivariate outliers were effectively identified, and visualizations were generated to facilitate clear communication of the results.